Konus, silindr, prizma va piramida

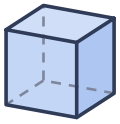
Qand bo'lagi — kub, gugurt qutisi — kuboid. Atrofimizda konus, silindr, prizma va piramida shaklidagi narsalar ham ko'p. Ularni asoslari va yoqlariga qarab bir-biridan ajratamiz.

1-QISM

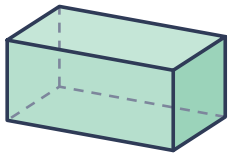
Mavzu

ESLAB OLAYLIK

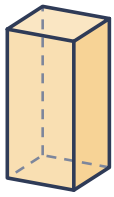
- Jismni o'rab turgan tekis bo'laklar uning **yoqlari** deyiladi.
- Kubning 6 ta yog'i bor, hammasi — teng kvadratlar. Masalan, qand bo'lagi.
- Kuboidning ham 6 ta yog'i bor: yo 6 tasi ham to'g'ri to'rtburchak, yo 4 tasi to'g'ri to'rtburchak va 2 tasi kvadrat. Masalan, gugurt qutisi.



Kub: 6 ta kvadrat yoq



Kuboid: 6 ta to'g'ri to'rtburchak yoq



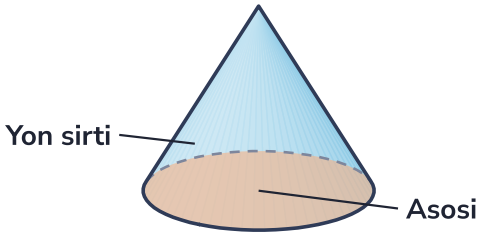
Kuboid: ustki va pastki yog'i — kvadrat, qolgan 4 tasi — to'g'ri to'rtburchak

Ba'zi jismlarda **yassi** (tekis) qismlar bilan birga **egri** (bukilgan) sirt ham bo'ladi.

TA'RIF · KONUS

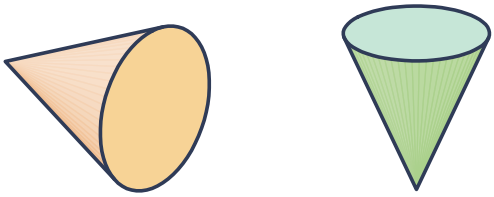
Konus bitta yassi **asos** va egri **yon sirt**dan iborat.

1-MISOL · KONUSNING QISMLARI



1. Asosi — doira. U yassi: konusni asosi bilan stolga qo'ysak, qimirlamay turadi.
2. Yon sirti — egri. U yuqoriga torayib, bitta nuqtada tugaydi.

2-MISOL · YOTGAN VA TESKARI KONUS



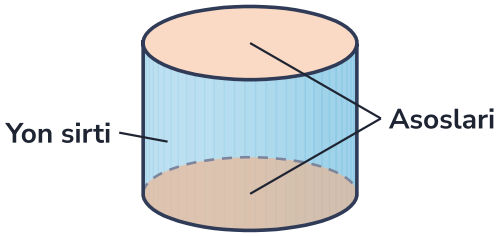
yotgan teskari

Ikkalasida ham bitta yassi asos va egri yon sirt bor — demak, ikkalasi ham konus. Voronkaning keng qismi, qalamning uchlangan qismi konus shaklida.

TA'RIF · SILINDR

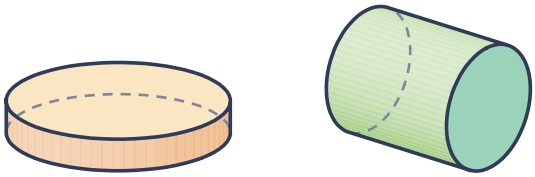
Silindr ikkita yassi **asos** va egri **yon sirt**dan iborat.

3-MISOL · SILINDRNING QISMLARI



1. Asoslari — ikkita teng doira: biri pastda, biri tepada. Ikkalasi ham yassi.
2. Yon sirti — egri. U ikki asosni o'rab turadi.

4-MISOL · PAST VA YOTGAN SILINDR



tanga g'o'la

Tanga juda past bo'lsa ham silindr: ikkita yassi asosi bor, yon sirti — ingichka egri hoshiya. Yotgan yog'och g'o'la, batareyka ham silindr shaklida.

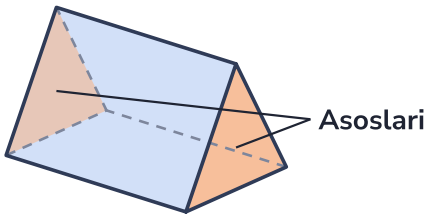


Shamni stolga yotqizsam, dumalab ketadi — yon sirti egri. Gugurt qutisi esa dumalamaydi: uning hamma yoqlari yassi.

TA'RIF · PRIZMA

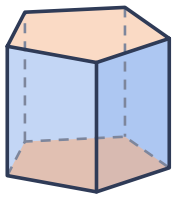
Prizmaning ikki **asosi** teng ko'pburchaklardan iborat. Prizmaning barcha yoqlari yassi bo'ladi.

5-MISOL · ASOSLARI UCHBURCHAK

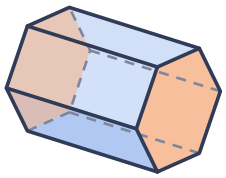


1. Ikki asosi — teng uchburchaklar (bo'yalgan). Bu prizma yon yog'i bilan yotibdi, shuning uchun asoslari old va orqa tomonda.
2. Qolgan 3 ta yog'i — to'g'ri to'rtburchaklar. Hamma yoqlari yassi.
3. Jami yoqlari: $2 + 3 = 5$ ta. Uchburchak devorli chodir shu shaklda.

6-MISOL · ASOSLARI BOSHQA KO'PBURCHAK



beshburchak



oltiburchak

1. Bo'yalgan asoslari: chapda — beshburchak, o'ngda — oltiburchak.
2. Yon yoqlari — to'g'ri to'rtburchaklar: chapdagi prizmada 5 ta, o'ngdagisida 6 ta.

Kub va kuboidni ham shunday tekshiring: ikki asosi teng to'rtburchak, hamma yoqlari yassi. Demak, kub ham, kuboid ham prizma.

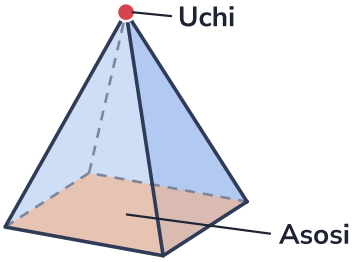


TA'RIF · PIRAMIDA

Piramida bitta yassi **asos** va uchburchaklardan iborat qiya **yoq**lardan iborat. Barcha uchburchaklarning umumiy uchi **piramidaning uchi** deb ataladi.

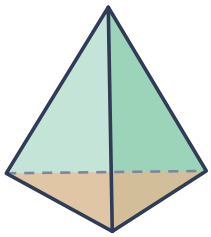
Piramidaning asosi istalgan ko'pburchak bo'lishi mumkin.

7-MISOL · PIRAMIDANING QISMLARI

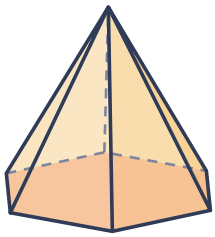


1. Asosi — kvadrat (bo'yalgan).
2. Qiya yoqlari — 4 ta uchburchak. Ular yuqorida bitta nuqtada — piramidaning uchida uchrashadi.
3. Jami yoqlari: $1 + 4 = 5$ ta.

8-MISOL · ASOSI BOSHQA KO'PBURCHAK



uchburchak



oltiburchak

Asosi	Qiya yoqlari	Jami
uchburchak	3	$1 + 3 = 4$
oltiburchak	6	$1 + 6 = 7$

Prizma va piramida o'xshash: ikkalasining hamma yoqlari yassi, asosi — ko'pburchak. Farqi: prizmaning asosi ikkita, yon yoqlari — to'rtburchaklar; piramidaning asosi bitta, yon yoqlari — uchburchaklar va ular bitta uchda uchrashadi.

To'rtta jismni solishtiramiz:

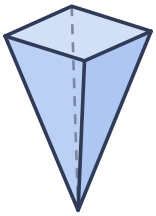
Jism	Asoslari	Yon tomoni
Konus	1 ta doira	egri sirt
Silindr	2 ta teng doira	egri sirt
Prizma	2 ta teng ko'pburchak	yassi to'rtburchak yoqlar
Piramida	1 ta ko'pburchak	yassi uchburchak yoqlar

2-QISM

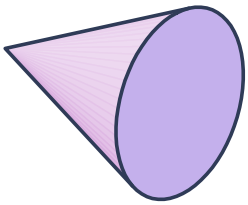
Misollar ishlash

1-MISOL · RASMGA QARAB NOMLASH

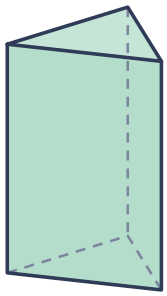
Jismlarni nomlang.



a)



b)



c)

1. Avval egri sirt bormi, qaraymiz. Bor bo'lsa — konus yoki silindr. Yo'q bo'lsa — prizma yoki piramida.
2. Keyin asoslarni sanaymiz. Bitta asos — konus yoki piramida. Ikkita teng asos — silindr yoki prizma.

a. Egri sirt yo'q. Asosi bitta — kvadrat (tepada), 4 ta uchburchak yoq pastda bitta uchda uchrashadi. Bu — **piramida**, u teskari turibdi.

b. Yon sirti egri, yassi asosi bitta (o'ngda). Bu — **konus**.

c. Egri sirt yo'q. Ustki va pastki yog'i — teng uchburchaklar, qolganlari to'g'ri to'rtburchaklar. Bu — **prizma**.

2-MISOL · BUYUM QAYSI JISM SHAKLIDA?

a) olti qirrali qalam (uchlanmagan); b) minora tomi: tagi kvadrat, to'rtta qiya uchburchak bo'lagi tepada tutashgan; c) pista solingan qog'oz kulyok.

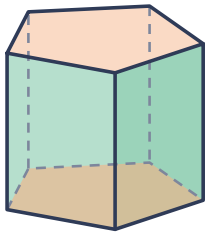
a. Qalamning ikki uchi — teng oltiburchaklar, qolgan yoqlari yassi. Bu — prizma, asoslari oltiburchak. Agar qalam dumaloq bo'lsa, yon sirti egri bo'ladi — u holda silindr.

b. Bitta asos (kvadrat) va bitta uchda uchrashgan uchburchak yoqlar — piramida.

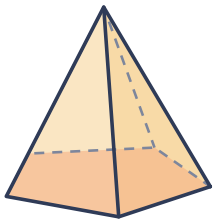
c. Kulyokning og'zi — doira, devori egri va pastda bitta nuqtada tugaydi. Bu — teskari turgan konus.

3-MISOL · YOQLARNI SANASH

Asosi beshburchak bo'lgan prizma va piramidaning nechtadan yog'i bor?



prizma



piramida

1. Prizma: 2 ta asos va har bir asos tomoniga bittadan — 5 ta yon yoq. Jami $2 + 5 = 7$ ta.
2. Piramida: 1 ta asos va 5 ta uchburchak yoq. Jami $1 + 5 = 6$ ta.

Jismni boshqa tomoniga ag'darib qo'ysak ham, nomi o'zgarmaydi. Asosni pastdan emas, shaklidan qidiraman.



DIQQAT!

1. Asosi doira bo'lsa — konus, ko'pburchak bo'lsa — piramida: ✗ asosi doira, demak piramida ✓ **asosi doira — konus**
2. Yon sirti egri jism prizma bo'lmaydi: ✗ ikki asosi bor, demak prizma ✓ **ikki asosi bor, yon sirti egri — silindr**
3. Yotgan prizmaning pastki yog'i doim ham asos emas: ✗ asosi — pastdagi to'rtburchak ✓ **asoslari — ikkita teng uchburchak**
4. Yoqlarni sanaganda asoslarni unutmang: ✗ asoslari beshburchak — 5 ta yoq ✓ $2 + 5 = 7$ **ta yoq**
5. Kub va kuboid ham prizma: ✗ kub prizma emas ✓ **kub — asoslari kvadrat bo'lgan prizma**

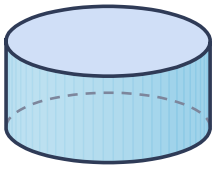
3-QISM

Mustahkamlash

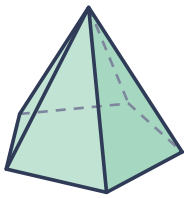
1. Jismning nomini yozing. Nechta yog'i kvadrat, nechtasi kvadrat bo'lmagan to'g'ri to'rtburchak?
   a) Qirrasi 5 cm bo'lgan kub. b) Bo'yi 6 cm, eni 2 cm, balandligi 4 cm bo'lgan jism. c) Bo'yi 3 cm, eni 3 cm, balandligi 8 cm bo'lgan jism.

2. Qaysi biri kub, qaysi biri kuboid shaklida?
   a) bulon kubigi; b) sharbat qutisi; c) Rubik kubigi; d) lug'at kitobi; e) harfli o'yinchoq kubik; f) muzlatgich.

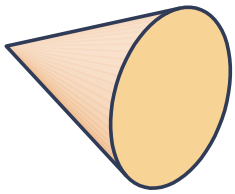
3. Rasmdagi jismlarni nomlang.



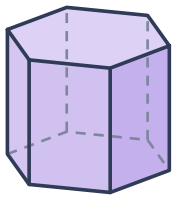
a)



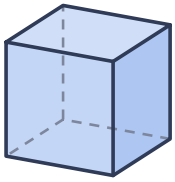
b)



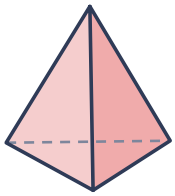
c)



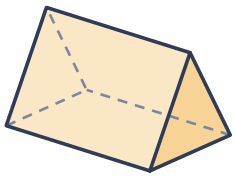
d)



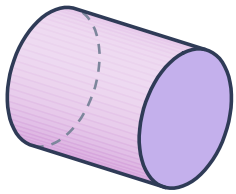
e)



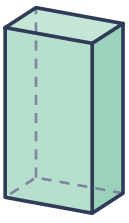
f)



g)



h)



i)

4. Jadvalni daftaringizga chizib to'ldiring.

Jism	Yassi asoslari soni	Yon tomoni: egri yoki yassi?
Silindr	?	?
Asosi to'rtburchak bo'lgan piramida	?	?
Asoslari beshburchak bo'lgan prizma	?	?
Konus	?	?

5. Buyum qaysi jism shaklida?
   a) yo'l konusi; b) uchburchak shaklidagi choy paketi (to'rtta yog'i ham uchburchak);
   c) sham; d) uchburchak shaklida kesilgan pishloq bo'lagi; e) baraban; f) muz kubigi.

6. Jismning nechta yog'i bor? Ular qanday shakllar?
   a) Asosi oltiburchak bo'lgan piramida. b) Asoslari uchburchak bo'lgan prizma. c) Asosi to'g'ri to'rtburchak bo'lgan piramida.

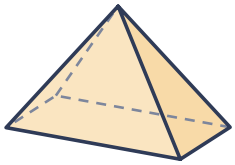
7. Prizma va piramidaning ikkita o'xshash va ikkita farqli tomonini yozing.

8. Men qaysi jismman?
   a) Ikkita yassi asosim bor, yon sirtim egri. b) Bitta asosim va to'rtta uchburchak yog'im bor. c) 6 ta yog'imning hammasi teng kvadrat. d) Bitta yassi asosim bor, yon sirtim egri.

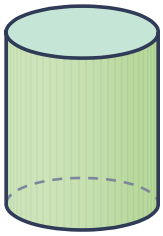
4-QISM

Mustaqil ish

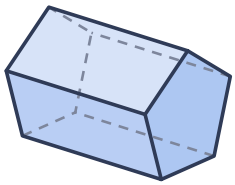
1. Rasmdagi jismlarni nomlang.



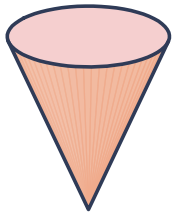
a)



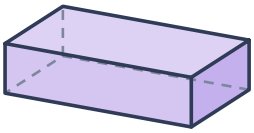
b)



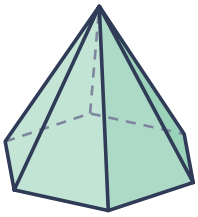
c)



d)



e)



f)

2. Uyingizdan yoki sinfxonadan konus, silindr, prizma va piramida shaklidagi bittadan narsa toping va nomini yozing.

3. Buyum qaysi jism shaklida?
   a) dumaloq tort; b) akvarium; c) tagi kvadrat, to'rtta uchburchak devori tepada bitta ustunga yig'ilgan chodir; d) qurilishdagi qum uyumi.

4. To'g'rimi yoki noto'g'rimi? Noto'g'ri bo'lsa, tuzating.
   a) Konusni teskari qo'ysak, u boshqa jismga aylanadi.
   b) Asoslari oltiburchak bo'lgan prizmaning hamma yoqlari yassi.
   c) Yotgan prizmaning stolga tegib turgan yog'i doim uning asosi bo'ladi.

5. **Xatoni toping.** Diyor batareykaga qarab: «Ikkita yassi asosi bor, demak bu prizma», — dedi. Xato nimada?

6. **Qiyinroq.** Qaysi biri ortiqcha? Nega?
   a) kub, kuboid, silindr, asoslari oltiburchak bo'lgan prizma.
   b) silindr, prizma, konus, kuboid.

5-QISM

Javoblar

Mustahkamlash

1-masala

a) Kub: 6 ta yog'i kvadrat. b) Kuboid: kvadrat yog'i yo'q, 6 ta to'g'ri to'rtburchak (6×2, 6×4, 2×4 — har biridan 2 tadan). c) Kuboid: ustki va pastki yog'i — 3×3 kvadrat (2 ta), qolgan 4 tasi — 3×8 to'g'ri to'rtburchak.

2-masala

Kub: a), c), e). Kuboid: b), d), f).

3-masala

a) silindr; b) piramida (asosi beshburchak); c) konus; d) prizma (asoslari oltiburchak); e) kub; f) piramida (asosi uchburchak); g) prizma (asoslari uchburchak); h) silindr; i) kuboid. Kub va kuboid — prizma ham.

4-masala

Silindr: 2 ta, egri. Piramida: 1 ta, yassi (4 ta uchburchak). Prizma: 2 ta, yassi (5 ta to'rtburchak). Konus: 1 ta, egri.

5-masala

a) konus; b) piramida (asosi uchburchak); c) silindr; d) prizma (asoslari uchburchak); e) silindr; f) kub.

6-masala

a) $1 + 6 = 7$ ta: 1 ta oltiburchak va 6 ta uchburchak. b) $2 + 3 = 5$ ta: 2 ta uchburchak va 3 ta to'g'ri to'rtburchak. c) $1 + 4 = 5$ ta: 1 ta to'g'ri to'rtburchak va 4 ta uchburchak.

7-masala

O'xshash: hamma yoqlari yassi; asosi — ko'pburchak. Farqi: prizmaning asosi ikkita, piramidaniki bitta; prizmaning yon yoqlari to'rtburchak, piramidaniki uchburchak; piramidaning uchi bor.

8-masala

a) silindr; b) piramida (asosi to'rtburchak); c) kub; d) konus.

Mustaqil ish

1-masala

a) piramida (asosi to'g'ri to'rtburchak); b) silindr; c) prizma (asoslari beshburchak); d) konus (teskari turibdi); e) kuboid; f) piramida (asosi oltiburchak).

2-masala

Javoblar har xil bo'ladi. Masalan: konus — voronka; silindr — sham, batareyka; prizma — olti qirrali qalam; piramida — uchburchak choy paketi.

3-masala

a) silindr; b) kuboid; c) piramida (asosi kvadrat); d) konus.

4-masala

a) Noto'g'ri: teskari turgan konus ham konus — bitta yassi asosi va egri yon sirti bor. b) To'g'ri: prizmaning barcha yoqlari yassi. c) Noto'g'ri: masalan, yon yog'i bilan yotgan prizmaning asoslari — ikki uchidagi uchburchaklar.

5-masala

Batareykaning yon sirti egri. Prizmaning esa hamma yoqlari yassi bo'ladi. Ikki yassi asos va egri yon sirt — bu silindr.

6-masala

a) Silindr: qolganlari prizma, hamma yoqlari yassi; silindrning yon sirti egri. b) Konus: uning asosi bitta, qolganlarining asosi ikkita.